voc triage one pager

VOC-Triage: AI for Disease-Associated Body Scent Biomarkers

One-Liner

VOC-Triage is an interpretable AI research platform that analyzes body-emitted volatile organic compound (VOC) profiles to help researchers identify disease-associated chemical signatures and prioritize samples for further clinical investigation.

Core Idea

Many diseases can alter metabolism, inflammation, microbiome activity, lipid oxidation, and other biological pathways. These changes can affect the volatile chemicals released through breath, sebum, skin, sweat, urine, or stool. Instead of treating smell as subjective, VOC-Triage treats the body's scent chemistry as biomedical data.

The platform does **not** diagnose disease. It acts as a **VOC-native triage layer**: an early, non-invasive research signal that helps scientists decide which samples, patients, or cohorts should be prioritized for deeper validation.

Problem

Researchers studying cancer, neurodegenerative disease, inflammation, and metabolic disorders are increasingly interested in VOC biomarkers, but the field is hard to translate because:

- VOC studies are scattered across different papers, instruments, diseases, and sample types
- chemical signatures are often messy, inconsistent, and hard to interpret
- a single VOC may appear in multiple diseases or non-disease contexts
- researchers need biomarker panels, not one "magic molecule"
- ML classifiers are often black boxes and do not explain which features matter
- designing validation studies requires knowing confounders like smoking, diet, infection, medication, or inflammation

The real pain is not "can people smell disease?" The pain is:

Researchers have complex VOC and metabolomics data, but need help turning it into interpretable biomarker hypotheses.

Solution

VOC-Triage helps researchers move from **literature chaos and raw chemical data** to **testable biomarker panels and validation plans**.

The platform can:

1. analyze VOC/metabolomics profiles
2. classify or triage samples when labeled datasets exist
3. highlight top chemical features driving a prediction
4. map VOCs to disease contexts, sample sources, and confounders
5. generate candidate biomarker panels
6. suggest follow-up validation experiments

Target User

The target user is **not a patient**. For the QBI hackathon, the target user is:

- UCSF cancer researchers studying early detection or tumor metabolism
- translational medicine and biomarker labs
- biotech diagnostic teams building non-invasive screening tools
- analytical chemistry or metabolomics groups working with biological samples

These users care about which VOCs have been reported, whether those VOCs are specific, what sample type is best, what confounders exist, and which chemical panels are worth validating.

MVP Features

1. VOC Triage Dashboard

A researcher selects or uploads a VOC/metabolomics sample. The platform outputs:

- triage score
- risk/priority category
- top chemical features
- feature importance
- possible biological interpretation
- caution/confounder notes

Example output:

Triage Flag: elevated follow-up priority
Possible Signal: aldehyde and ketone cluster abnormality

Relevant Contexts: lung cancer VOC studies, oxidative stress, smoking-related confounders
Recommendation: not diagnostic; prioritize for confirmatory testing or validation cohort inclusion

2. VOC Disease Atlas

A searchable map of disease-associated VOC evidence.

Search examples:

- lung cancer
- colorectal cancer
- Parkinson's disease
- diabetes
- inflammation

Each disease card shows:

- reported VOCs or chemical families
- sample source: breath, stool, urine, skin, sebum
- evidence level
- possible pathways
- major confounders
- whether a public dataset/model exists

3. Specificity Checker

Researchers can search a VOC and see whether it is disease-specific or commonly confounded.

Example:

VOC: acetone
Appears in: diabetes/metabolism, fasting/ketosis, some cancer VOC studies
Specificity: low
Recommendation: weak standalone biomarker; possible panel feature

4. Biomarker Panel Builder

For a disease like lung or colorectal cancer, the tool suggests a candidate VOC panel and validation strategy.

Example:

Lung Cancer Candidate Panel:

- aldehyde cluster
- ketone cluster
- alkane/lipid peroxidation cluster
- aromatic/smoking-confounder cluster

Validation Plan:

- compare lung cancer vs healthy controls
- include COPD and smoker controls
- control for diet, medication, and environment
- evaluate sensitivity, specificity, and panel robustness

Proof-of-Concept Dataset Strategy

The strongest hackathon path is to use **Parkinson's sebum data** as the proof-of-concept ML dataset because public sebum datasets already exist and the Joy Milne story provides a powerful olfaction-based origin. The platform can then expand toward cancer VOC triage as the broader use case.

Recommended scope:

- **Main ML demo:** Parkinson's sebum classifier using public metabolomics/VOC data
- **Expansion layer:** lung cancer and colorectal cancer VOC atlas
- **Research value:** show how the same platform can support cancer biomarker discovery once clean cancer VOC datasets are available

Why This Solves for UCSF / QBI

UCSF and QBI care about biomedical discovery, translational research, interpretable AI, and tools that help scientists move from raw data to actionable hypotheses. VOC-Triage fits because it supports:

- early disease biomarker discovery
- non-invasive screening research
- risk stratification
- interpretable ML for chemical datasets
- experiment design and validation planning
- cancer and neurodegenerative disease research

This is not a consumer wellness app. It is a research tool for helping scientists build and validate VOC-based triage models.

Positioning Statement

VOC-Triage uses body-emitted chemical signatures as an early, non-invasive research signal to prioritize samples, patients, or cohorts for further testing. It is not a diagnostic tool; it is a biomarker discovery and hypothesis-generation platform for translational medicine.

Best Pitch Summary

Current disease screening often depends on expensive, invasive, or late-stage tests. But many diseases alter metabolism, inflammation, microbiome activity, and lipid oxidation, which can change the volatile compounds emitted through breath, urine, stool, skin, or sebum.

VOC-Triage is an AI research platform that uses these body-emitted chemical profiles as a non-invasive pre-screening layer. It helps researchers identify biomarker-like patterns, rank follow-up priority, explain which chemical features matter, and design validation studies.

We start with Parkinson's sebum data because it has public datasets and a strong olfaction-based origin story, then expand the platform toward cancer VOC triage for lung and colorectal cancer.

feature set pure gen

VOC-Triage Feature Set + User Flow

Product Purpose

VOC-Triage is an AI research platform that helps biomedical researchers analyze body-emitted volatile organic compound (VOC) data and turn it into interpretable biomarker hypotheses.

The platform is designed for researchers working on non-invasive disease screening, cancer biomarker discovery, neurodegenerative disease, metabolomics, breathomics, sebum analysis, and translational medicine.

VOC-Triage does not diagnose disease. It helps researchers identify chemical patterns, evaluate biomarker specificity, and decide which samples or biomarker panels should be prioritized for further validation.

Core User

Primary User

A biomedical researcher, cancer researcher, translational medicine researcher, metabolomics analyst, or biotech biomarker team.

What They Need

The user wants to answer questions like:

- Which VOCs are associated with a disease?
- Which VOCs are specific versus heavily confounded?
- Which chemical features separate disease samples from controls?
- Which sample type is most promising: breath, urine, stool, skin, sebum, or sweat?
- Which biomarker panel should be validated next?
- What follow-up experiment should be designed?

Feature Set

1. VOC Disease Atlas

Purpose

A searchable evidence map that connects diseases to reported VOCs, sample sources, chemical classes, confounders, and research evidence.

What the user can search

- lung cancer
- colorectal cancer
- breast cancer
- Parkinson’s disease
- diabetes
- inflammation
- infection
- oxidative stress

What the feature shows

For each disease, the atlas displays:

- reported VOCs or chemical families
- sample source: breath, urine, stool, skin, sweat, sebum
- evidence level: strong, moderate, emerging, weak
- possible biological pathways
- known confounders
- related papers or sources
- whether a usable dataset exists
- whether a model has been trained

Example Output

Disease: Lung cancer
Sample Source: breath
Reported Chemical Families: aldehydes, ketones, alkanes, aromatics
Possible Pathways: oxidative stress, lipid peroxidation, tumor metabolism
Major Confounders: smoking, COPD, diet, medication, environment
Research Use: candidate panel design and validation planning

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2. Sample Triage Dashboard

Purpose

Lets a researcher select or upload a VOC/metabolomics sample and receive an interpretable triage-style output.

Input

The user provides:

- a sample profile
- uploaded VOC/metabolomics file
- or a preloaded demo sample from a public dataset

Output

The dashboard displays:

- triage flag
- priority score
- possible disease-associated pattern
- confidence level
- top chemical features
- feature importance chart
- possible confounders
- recommended follow-up step

Example Output

Sample ID: A042
Triage Flag: elevated follow-up priority
Priority Score: 78/100
Possible Signal: aldehyde and ketone cluster abnormality
Relevant Contexts: lung cancer VOC studies, oxidative stress, smoking-related confounders
Recommended Next Step: not diagnostic; prioritize for confirmatory testing or validation cohort inclusion

Important Framing

The output should never say:

> This patient has cancer.

It should say:

> This sample contains VOC patterns that may warrant further investigation.

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3. Biomarker Panel Builder

Purpose

Helps researchers move from individual VOCs to a candidate multi-VOC biomarker panel.

Why This Matters

Most diseases cannot be identified by one molecule. Researchers need panels that combine multiple chemical features and account for confounders.

User Flow

The user selects a disease, such as:

- lung cancer
- colorectal cancer
- Parkinson’s disease
- diabetes

The tool generates:

- candidate VOC panel
- chemical class grouping
- rationale for each feature group
- confounder warnings
- validation cohort recommendations
- suggested evaluation metrics

Example Output

Disease: Lung cancer

Candidate Panel:

- aldehyde cluster
- ketone cluster
- alkane/lipid peroxidation cluster
- aromatic/smoking-confounder cluster

Validation Plan:

- compare lung cancer samples against healthy controls
- include COPD controls
- include smoker and non-smoker groups
- control for diet, medication, and environmental exposure
- evaluate sensitivity, specificity, ROC-AUC, and feature stability

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4. VOC Specificity Checker

Purpose

Lets researchers search one VOC and evaluate whether it is disease-specific or heavily confounded.

Input Examples

- acetone
- nonanal
- hexanal
- isoprene
- toluene
- eicosane
- benzaldehyde

What the feature shows

For each VOC, the tool displays:

- diseases where the VOC has been reported
- non-disease contexts where it appears
- sample sources
- chemical class
- specificity score
- confounder warnings
- whether it should be used alone or in a panel

Example Output

VOC: acetone
Chemical Class: ketone
Reported Contexts: diabetes/metabolism, fasting/ketosis, some cancer VOC studies

Specificity: low
Confounders: diet, fasting, metabolic state
Recommendation: weak standalone biomarker; possible supporting feature in a multi-VOC panel

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5. Explainability Layer

Purpose

Makes model predictions interpretable for researchers.

What it shows

- top features driving the triage score
- feature importance chart
- SHAP-style explanation if available
- chemical family grouping
- pathway-level interpretation
- model confidence
- evidence limitations

Example

Instead of only saying:

> Disease-associated pattern detected.

The app explains:

> The triage score was mainly driven by elevated aldehyde and hydrocarbon features, which may be associated with oxidative stress and lipid peroxidation. However, these features may also be affected by smoking, inflammation, or environmental exposure.

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6. Confounder Alert System

Purpose

Warns researchers when a VOC pattern may be influenced by non-disease factors.

Confounder categories

- smoking
- diet
- fasting
- medication
- infection
- inflammation
- microbiome changes
- environmental exposure
- sample storage
- age or sex differences
- instrument differences

Example

Warning: Several selected VOCs are also associated with smoking exposure and COPD. A lung cancer validation cohort should include smoker controls and COPD controls.

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7. Validation Plan Generator

Purpose

Helps researchers design the next experiment after identifying a candidate VOC panel.

Output

The generator suggests:

- comparison groups
- sample source
- control groups
- confounders to control
- metrics to evaluate
- recommended follow-up tests
- whether the biomarker panel is ready for validation or still exploratory

Example

Validation Plan for Colorectal Cancer VOC Panel:

- collect urine or stool VOC samples

- compare CRC patients vs healthy controls
- include inflammatory bowel disease controls
- control for diet, antibiotics, and microbiome differences
- evaluate sensitivity, specificity, ROC-AUC, and panel reproducibility
- test whether the panel performs better than individual VOCs

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8. Public Dataset Model Demo

Purpose

Provides the actual ML proof-of-concept for the hackathon.

Recommended MVP Dataset

Use a public Parkinson's sebum metabolomics/VOC dataset because it is more accessible and has a strong olfaction-based origin story.

What the demo model does

- trains a classifier on public sebum data
- predicts control vs Parkinson's or multi-class disease status if labels allow
- shows feature importance
- shows confusion matrix
- shows ROC curve if possible
- explains which biomarker clusters separate groups

Why This Works

Parkinson's is the proof-of-concept dataset, but the platform vision expands toward cancer VOC triage for lung and colorectal cancer.

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9. Research Export

Purpose

Lets researchers export results into a usable format.

Export includes

- disease or VOC searched
- candidate panel
- top features
- specificity notes
- confounders
- validation plan
- disclaimer
- citations or data source notes

Export formats

MVP version:

- copyable text
- downloadable PDF
- downloadable CSV of selected VOCs

Full User Flow

Flow 1: Researcher Searches a Disease

Step 1: User opens VOC-Triage

The homepage explains:

> VOC-Triage helps researchers explore disease-associated body VOC patterns and generate biomarker hypotheses for non-invasive screening research.

Step 2: User searches “lung cancer”

The app opens the Lung Cancer Disease Card.

Step 3: User reviews evidence

The card shows:

- reported VOC classes
- sample source: breath
- possible pathways
- evidence level

- confounders
- related diseases

Step 4: User clicks “Build Biomarker Panel”

The app suggests a multi-VOC candidate panel grouped by chemical families.

Step 5: User checks specificity

The app flags VOCs that are not specific enough or are heavily confounded by smoking, COPD, or environment.

Step 6: User generates validation plan

The app suggests control groups, sample collection method, and validation metrics.

Step 7: User exports the result

The researcher exports a candidate panel and validation plan.

Flow 2: Researcher Uploads or Selects a Sample

Step 1: User selects a demo dataset

For MVP, the user selects a Parkinson’s sebum dataset sample.

Step 2: App runs model

The classifier analyzes the chemical feature profile.

Step 3: App displays triage result

The dashboard shows:

- predicted class or triage flag
- confidence score
- top features
- feature importance
- possible biological interpretation

Step 4: App explains limitations

The tool states:

> This is a research prototype, not a diagnostic output.

Step 5: User reviews follow-up recommendation

The app recommends whether this sample should be included in further validation or compared against specific controls.

Flow 3: Researcher Searches a VOC

Step 1: User searches “acetone”

The VOC Specificity Checker opens.

Step 2: App shows where acetone appears

The app lists:

- diabetes/metabolism
- fasting/ketosis
- possible cancer VOC studies
- breath and skin contexts

Step 3: App gives specificity score

The app labels acetone as low-specificity.

Step 4: App gives recommendation

The app says:

> Acetone should not be used as a standalone disease marker, but may contribute to a broader metabolic or cancer-associated VOC panel.

MVP Build Priorities

Must-Have

1. homepage with clear research framing
2. disease search
3. disease VOC cards
4. specificity checker
5. biomarker panel builder
6. demo sample triage dashboard
7. simple ML proof-of-concept or simulated demo clearly marked as prototype
8. disclaimer that the tool is not diagnostic

Should-Have

1. feature importance chart
2. confounder warning system
3. exportable validation plan
4. disease comparison view
5. citations or evidence source labels

Nice-to-Have

1. full model training pipeline
2. SHAP explainability
3. UMAP/PCA visualization
4. multi-disease dataset support
5. PDF report generation
6. researcher collaboration notes

Demo Flow for Judges

Demo Script

1. Start with the problem:

   VOC research is promising for early disease screening, but researchers struggle with scattered evidence, messy datasets, confounders, and black-box models.

2. Search "lung cancer."

Show the disease VOC card, reported chemical classes, sample source, and confounders.

3. Click “Build Biomarker Panel.”

Show a candidate panel and validation plan.

4. Search “acetone.”

Show that acetone is low-specificity and confounded by metabolism/fasting, proving the tool is scientifically cautious.

5. Open the demo classifier.

Show a Parkinson’s sebum sample being analyzed with a triage flag, top features, and explainability chart.

6. End with the platform vision:

> We start with Parkinson’s sebum data as a proof-of-concept, then expand to cancer VOC triage for lung and colorectal cancer as cleaner datasets become available.

Success Criteria

By the end of the hackathon, VOC-Triage should prove that:

1. VOCs can be organized into disease-specific research maps
2. researchers can evaluate specificity and confounders
3. interpretable AI can help prioritize candidate biomarker panels
4. the tool can support validation planning
5. the platform can connect olfaction science to serious biomedical research